

```
In [2]: import pandas as pd
```

```
In [3]: url="Student Mental health.csv"
```

```
In [4]: df=pd.read_csv(url)
```

```
In [5]: df
```

Out[5]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?
0	8/7/2020 12:02	Female	18.0	Engineering	year 1	3.00 - 3.49	No	Yes	No
1	8/7/2020 12:04	Male	21.0	Islamic education	year 2	3.00 - 3.49	No	No	Yes
2	8/7/2020 12:05	Male	19.0	BIT	Year 1	3.00 - 3.49	No	Yes	Yes
3	8/7/2020 12:06	Female	22.0	Laws	year 3	3.00 - 3.49	Yes	Yes	No
4	8/7/2020 12:13	Male	23.0	Mathemathics	year 4	3.00 - 3.49	No	No	No
...	...	...	...	...	...	...	...	...	...
96	13/07/2020 19:56:49	Female	21.0	BCS	year 1	3.50 - 4.00	No	No	Yes
97	13/07/2020 21:21:42	Male	18.0	Engineering	Year 2	3.00 - 3.49	No	Yes	Yes
98	13/07/2020 21:22:56	Female	19.0	Nursing	Year 3	3.50 - 4.00	Yes	Yes	No
99	13/07/2020 21:23:57	Female	23.0	Pendidikan Islam	year 4	3.50 - 4.00	No	No	No
100	18/07/2020 20:16:21	Male	20.0	Biomedical science	Year 2	3.00 - 3.49	No	No	No

101 rows × 11 columns



In [6]: `df.head(5)`

Out[6]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?	Do you have Panic attack?
0	8/7/2020 12:02	Female	18.0	Engineering	year 1	3.00 - 3.49	No	Yes	No	
1	8/7/2020 12:04	Male	21.0	Islamic education	year 2	3.00 - 3.49	No	No	Yes	
2	8/7/2020 12:05	Male	19.0	BIT	Year 1	3.00 - 3.49	No	Yes	Yes	
3	8/7/2020 12:06	Female	22.0	Laws	year 3	3.00 - 3.49	Yes	Yes	No	
4	8/7/2020 12:13	Male	23.0	Mathematics	year 4	3.00 - 3.49	No	No	No	

In [7]: `df.tail(5)`

Out[7]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?	Do you have Panic attack?
96	13/07/2020 19:56:49	Female	21.0	BCS	year 1	3.50 - 4.00	No	No	Yes	
97	13/07/2020 21:21:42	Male	18.0	Engineering	Year 2	3.00 - 3.49	No	Yes	Yes	
98	13/07/2020 21:22:56	Female	19.0	Nursing	Year 3	3.50 - 4.00	Yes	Yes	No	
99	13/07/2020 21:23:57	Female	23.0	Pendidikan Islam	year 4	3.50 - 4.00	No	No	No	
100	18/07/2020 20:16:21	Male	20.0	Biomedical science	Year 2	3.00 - 3.49	No	No	No	

In [8]: `df.shape`

Out[8]: (101, 11)

In [9]: `df.columns`

Out[9]: Index(['Timestamp', 'Choose your gender', 'Age', 'What is your course?', 'Your current year of Study', 'What is your CGPA?', 'Marital status', 'Do you have Depression?', 'Do you have Anxiety?', 'Do you have Panic attack?', 'Did you seek any specialist for a treatment?'], dtype='object')

In [10]: df.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 101 entries, 0 to 100
Data columns (total 11 columns):
#   Column                                Non-Null Count  Dtype
---  -
0   Timestamp                            101 non-null    object
1   Choose your gender                   101 non-null    object
2   Age                                  100 non-null    float64
3   What is your course?                101 non-null    object
4   Your current year of Study          101 non-null    object
5   What is your CGPA?                 101 non-null    object
6   Marital status                     101 non-null    object
7   Do you have Depression?             101 non-null    object
8   Do you have Anxiety?               101 non-null    object
9   Do you have Panic attack?         101 non-null    object
10  Did you seek any specialist for a treatment? 101 non-null    object
dtypes: float64(1), object(10)
memory usage: 4.8+ KB
```

In [11]: df.describe()

Out[11]:

	Age
count	100.00000
mean	20.53000
std	2.49628
min	18.00000
25%	18.00000
50%	19.00000
75%	23.00000
max	24.00000

In [12]: df["Age"]

Out[12]:

0	18.0
1	21.0
2	19.0
3	22.0
4	23.0
	...
96	21.0
97	18.0
98	19.0
99	23.0
100	20.0

Name: Age, Length: 101, dtype: float64

In [13]: `df[["Choose your gender", "What is your CGPA?"]]`

Out[13]:

	Choose your gender	What is your CGPA?
0	Female	3.00 - 3.49
1	Male	3.00 - 3.49
2	Male	3.00 - 3.49
3	Female	3.00 - 3.49
4	Male	3.00 - 3.49
...	...	...
96	Female	3.50 - 4.00
97	Male	3.00 - 3.49
98	Female	3.50 - 4.00
99	Female	3.50 - 4.00
100	Male	3.00 - 3.49

101 rows × 2 columns

In [14]: `df.iloc[0:5]`

Out[14]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?	Do you have F att
0	8/7/2020 12:02	Female	18.0	Engineering	year 1	3.00 - 3.49	No	Yes	No	
1	8/7/2020 12:04	Male	21.0	Islamic education	year 2	3.00 - 3.49	No	No	Yes	
2	8/7/2020 12:05	Male	19.0	BIT	Year 1	3.00 - 3.49	No	Yes	Yes	
3	8/7/2020 12:06	Female	22.0	Laws	year 3	3.00 - 3.49	Yes	Yes	No	
4	8/7/2020 12:13	Male	23.0	Mathematics	year 4	3.00 - 3.49	No	No	No	

In [15]: `df.iloc[0:5,0:5]`

Out[15]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study
0	8/7/2020 12:02	Female	18.0	Engineering	year 1
1	8/7/2020 12:04	Male	21.0	Islamic education	year 2
2	8/7/2020 12:05	Male	19.0	BIT	Year 1
3	8/7/2020 12:06	Female	22.0	Laws	year 3
4	8/7/2020 12:13	Male	23.0	Mathematics	year 4

In [16]: `df.isnull()`

Out[16]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?	Do you have a mental health problem?
0	False	False	False	False	False	False	False	False	False	False
1	False	False	False	False	False	False	False	False	False	False
2	False	False	False	False	False	False	False	False	False	False
3	False	False	False	False	False	False	False	False	False	False
4	False	False	False	False	False	False	False	False	False	False
...	...	...	...	...	...	...	...	...	...	...
96	False	False	False	False	False	False	False	False	False	False
97	False	False	False	False	False	False	False	False	False	False
98	False	False	False	False	False	False	False	False	False	False
99	False	False	False	False	False	False	False	False	False	False
100	False	False	False	False	False	False	False	False	False	False

101 rows × 11 columns

In [21]: `from sklearn import preprocessing`

In [22]: `df.head()`

Out[22]:

	Timestamp	Choose your gender	Age	What is your course?	Your current year of Study	What is your CGPA?	Marital status	Do you have Depression?	Do you have Anxiety?	Do you have a mental health problem?
0	8/7/2020 12:02	Female	18.0	Engineering	year 1	3.00 - 3.49	No	Yes	No	
1	8/7/2020 12:04	Male	21.0	Islamic education	year 2	3.00 - 3.49	No	No	Yes	
2	8/7/2020 12:05	Male	19.0	BIT	Year 1	3.00 - 3.49	No	Yes	Yes	
3	8/7/2020 12:06	Female	22.0	Laws	year 3	3.00 - 3.49	Yes	Yes	No	
4	8/7/2020 12:13	Male	23.0	Mathematics	year 4	3.00 - 3.49	No	No	No	

In [23]: `min_max_scaler=preprocessing.MinMaxScaler()`

In [24]: `x=df.iloc[:,4]`

In [31]: `print(x.dtypes)`

```
What is your course?    object
dtype: object
```

In [36]: `x = pd.get_dummies(x, drop_first=True)`
`x_scaled = min_max_scaler.fit_transform(x)`

In [46]: `df_normalized=pd.DataFrame(x_scaled)`

In [47]: `df_normalized`

Out[47]:

	0	1	2	3	4	5	6	7	8	9	...	38	39	40	41	42	43	44	45	46
0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
2	0.0	0.0	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
3	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
4	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
96	0.0	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
97	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
98	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
99	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	...	1.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
100	0.0	0.0	0.0	0.0	0.0	0.0	1.0	0.0	0.0	0.0	...	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

101 rows × 48 columns



In [48]: `df['Choose your gender'].unique()`

Out[48]: `array(['Female', 'Male'], dtype=object)`

In [49]: `label_encoder=preprocessing.LabelEncoder()`

In [51]: `df['Choose your gender']=label_encoder.fit_transform(df['Choose your gender'])`

In [52]: `df['Choose your gender'].unique()`

Out[52]: `array([0, 1])`